Is photoperiod the dominant or least important cue for woody plant spring phenology?

Woody plant leafout and flowering each spring is strongly driven by warm spring temperatures, and thus has shifted days to weeks earlier with anthropogenic climate change. But how long this advance can continue depends on the relative importance of the two other major environmental cues—chilling (via the accumulation of cool winter temperatures) and photoperiod. Over a decade of focused research (e.g., Laube *et al.*, 2014; Ettinger *et al.*, 2020) has found chilling produces the largest shifts in spring phenology, often 3-5X larger than the response to photoperiod (Morales-Castilla *et al.*, 2024). But this research generally tests photoperiod responses in only one way (Fig. 1) that is disconnected from the molecular concept of critical photoperiod. At the same time, new studies suggest a dominating role of solstice mid-season phenology (Zohner *et al.*, 2023; Journé *et al.*, 2024), and a remarkably stable photoperiod response across most species (Fig. 2). Together these new results suggest an important and poorly understood role of photoperiod that may not be well tested by current methods. Building a mechanistic predictive model of spring phenology thus requires re-examining how much we actually know currently and developing new approaches.

Here, I want to explore:

1. How do current findings in woody plant spring phenology experiments compare to our understanding of photoperiod from the molecular literature? Most studies manipulate photoperiod alongside forcing and chilling temperatures usually by using two different constant photoperiods with only small responses (Fig. 1).

2. What experiments could help better understand mechanistically how photoperiod affects spring phenology?

We have data on many woody plant spring phenology experiments and can explore:

1. Differences in photoperiod responses for budburst versus flowering (since most molecular research is on flowering, right?).

2. Budburst % versus timing responses to photoperiod... would need to back up in the OSPREE database some to do this, but would be good to also show for a subset of species (Fagus and Betula perhaps?)

3. Whether any studies do light interruption (I have not found any in our database of woody plant spring phenology, but I saw some for strawberry, which was included by accident). That's lieten97 ... but I think we did a poor job of documenting this paper, maybe it is: `https://www.actahort.org/books/439/439_105.htm`

4. Compare findings we see from molecular experiments such as those that Stacey Harmer pointed out where the response is considered not important, but is actually just small compared to the larger responses seen usually to critical photoperiod.

Thinking in 2026.... Other stuff we could add...

1. Dan's stuff on male vs female flowers

2. Dan's stuff on how photoperiod and thermoperiodicity often covaried

3. Korner work on leafout is photoperiod (cell elongation/photosynthesis) while budburst is a trigger (more like the flowering literature)

4. Deirdre's work on traits (photoperiod matters)

5. Photoperiod and budset (but Rob Guy actually did photoeperiod interruption experiments)

6. Better models of forcing (log-log or threshold ones) and what we find in photoperiod effects then...

7. Probably need to give a REALLY concrete example of why knowing the mechanism matters.

Be also sure to review what we say in Nacho's 2024 paper about photoperiod and phylogeny as it's good. We can just repeat some of this....

* write paper, then do any additional analyses; send to Dan, then ask Stacey Harmer to join it after quick draft? Maybe see if Justin wants to join, since he is a molecular expert!

Outline

1. Introduction
    (a) Spring events matter to plants, and people like to record them.
        i. Spring leafout and flowering are some of the most important events for plants
        ii. They're critical moments for growth and reproduction
        iii. They also happen in the spring, often an interesting moment balancing risk and reward
        iv. And a time when humans like to monitor life, as we all emerge from winter (US-NPN shows obs are spring?)
        v. Shifts in phenology, especially spring events, have become the most recorded biological response, and led to a new sub-literature on phenology alongside its place in fundamental plant biology: Sort of a global change phenology sub-discipline
    (b) Global change phenology exploded over the last 20 years, leading to tons of work on spring phenology, especially for woody plants.
        i. Woody plants are important for carbon models (and thus CC modeling)
        ii. Forecasting leafout is big, and lots of work on figuring it out.
        iii. There are three cues: chlling, forcing, photoperiod ...
        iv. One big question is how much each matters and how?
    (c) Photoperiod seems to be either the most important, or least important cue
        i. Lots of work a decade ago saying it mattered the most
        ii. Then people said it really did not
        iii. A recent meta-analysis said it has the smallest response, but the evolutionary signal in photoperiod was intriguing...
    (d) Paragraph on the evolutionary signal (we say this really well in OSPREE 2024 paper so review there)
    (e) This evolutionary signal suggests a conserved molecular basis to photoperiod, but why is the type of photoperiod response observed in woody plant phenology?
        i. There's not a lot of evidence of a critical photoperiod
        ii. So it's not the classic internal coincidence model
        iii. Maybe it's some other model...
        iv. Because actually new studies in molecular biology are finding new models ... especially under more natural conditions.

(f) Given daylength is so clearly important to fundamental plant biology, and getting it right in models of woody plant leafout has big carbon implications, figuring out what the hell is going on here seems important, so – here – we'll review what is going on, and suggest a path forward.

2. Current well-established models of photoperiod control in plant leafout and flowering...
   (a) Hourglass
   (b) Circadian Clock (internal vs. external coincidence model)
      i. External is one clock
      ii. Internal is TWO clocks
      iii. I think that both are out there, but internal is "more widely supported" in animals (Imaizumi)
3. Evidence in woody plant phenology (this section needs some subsections...)
   (a) Lots about critical photoperiods and stuff...
   (b) But studies never test for critical photoperiods
   (c) And if you look at the data, it's not so dramatic (Walde, OSPREE results)
   (d) But it's there! Nacho's OSPREE paper.... bridge evolution back to molecular ... and new models emerging in the molecular literature
4. New models of photoperiod
   (a) Photosynthesis matters
   (b) Imaizumi et al. 2025 (who is at the UW!) – overviews recent 'more natural' experiments that have turned up weird stuff → this is where phenological studies come in!
   (c) (How do we bridge these fields!? Or...) These results suggest a path forward to better understand the mechanisms of photoperiod control in woody plants
5. Finding the mechanism for photoperiod control in woody plants
   (a) The molecular people are starting to use natural experiments, they should keep doing that ...
   (b) Us climate change phenology people may need to think of some more types of experiments
      i. Some emphasis on realistic conditions (ramping etc.) which is good, but don't lose track of the value and aim of experiments
         A. Realistic experiments may help you forecast, but the big goal of experiments is to...
         B. .... identify effects, even if you need a sledgehammer to do it.

C. And experiments are critical to discover the mechanism behind these effects; we do NOT do enough of this.

ii. Woody phenology people need some experiments to help get at mechanism ... we should add...

A. Some boring basic stuff like Okie11 recommends, that includes;controlled photoperiod during all dormancy treatments (versus all dark)

B. Night-interruption (there's one at least done, let's do more!)

C. Really short photoperiods – to get at what is going in Nacho's results (but this is hard because then you reduce photosynthesis)

D. Potentially more like Gendron – separate out photosynthesis from actual triggers.

iii. Somewhere, we need to mention species diffs ...

A. Species diffs are there in woody plant phenology; just look at Fagus

B. Working from a phylogenetic perspective on several clades may be critical (Fagacaese and Betulaceae)

C. .. and thinking on bud differences (how much light gets through? Flower vs leaf buds, types etc.)

6. Wrap up? Say how exciting a time it is. Rah, rah.

References

Ettinger, A.K., Chamberlain, C.J., Morales-Castilla, I., Buonaiuto, D.M., Flynn, D.F.B., Savas, T., Samaha, J.A. & Wolkovich, E.M. (2020) Winter temperatures predominate in spring phenological responses to warming. *Nature Climate Change* **10**, 1137–U119.

Journé, V., Szymkowiak, J., Foest, J., Hacket-Pain, A., Kelly, D. & Bogdziewicz, M. (2024) Summer solstice orchestrates the subcontinental-scale synchrony of mast seeding. *Nature Plants* **10**, 367–373.

Laube, J., Sparks, T.H., Estrella, N., Hofler, J., Ankerst, D.P. & Menzel, A. (2014) Chilling outweighs photoperiod in preventing precocious spring development. *Global Change Biology* **20**, 170–182.

Morales-Castilla, I., Davies, T., Legault, G., Buonaiuto, D., Chamberlain, C.J., Ettinger, A.K., Garner, M., Jones, F.A., Loughnan, D., Pearse, W.D. *et al.* (2024) Phylogenetic estimates of species-level phenology improve ecological forecasting. *Nature Climate Change* pp. 1–7.

Zohner, C.M., Mirzagholi, L., Renner, S.S., Mo, L., Rebindaine, D., Bucher, R., Palouš, D., Vitasse, Y., Fu, Y.H., Stocker, B.D. *et al.* (2023) Effect of climate warming on the timing of autumn leaf senescence reverses after the summer solstice. *Science* **381**, eadf5098.

Figures

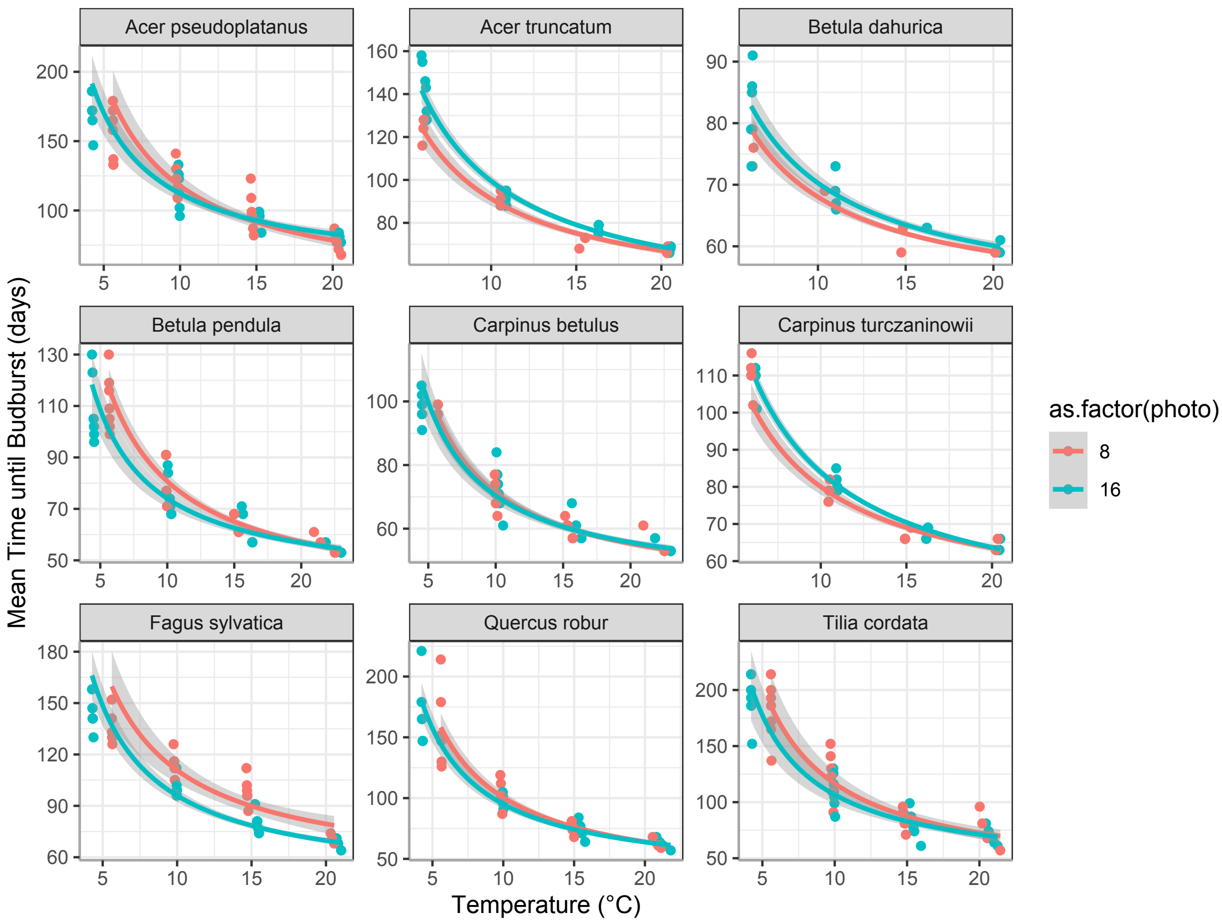

Figure 1: Results from Walde *et al.* 2023 for budburst timing across species under high chilling and two photoperiods: 8 and 16 hours. This type of study is the most common type done in woody plant phenology—it uses cuttings of dormant buds exposed to chilling (cool temperatures) followed by different combinations of warm temperatures and photoperiods (this study is unique though in covering a wide range of forcing temperatures).

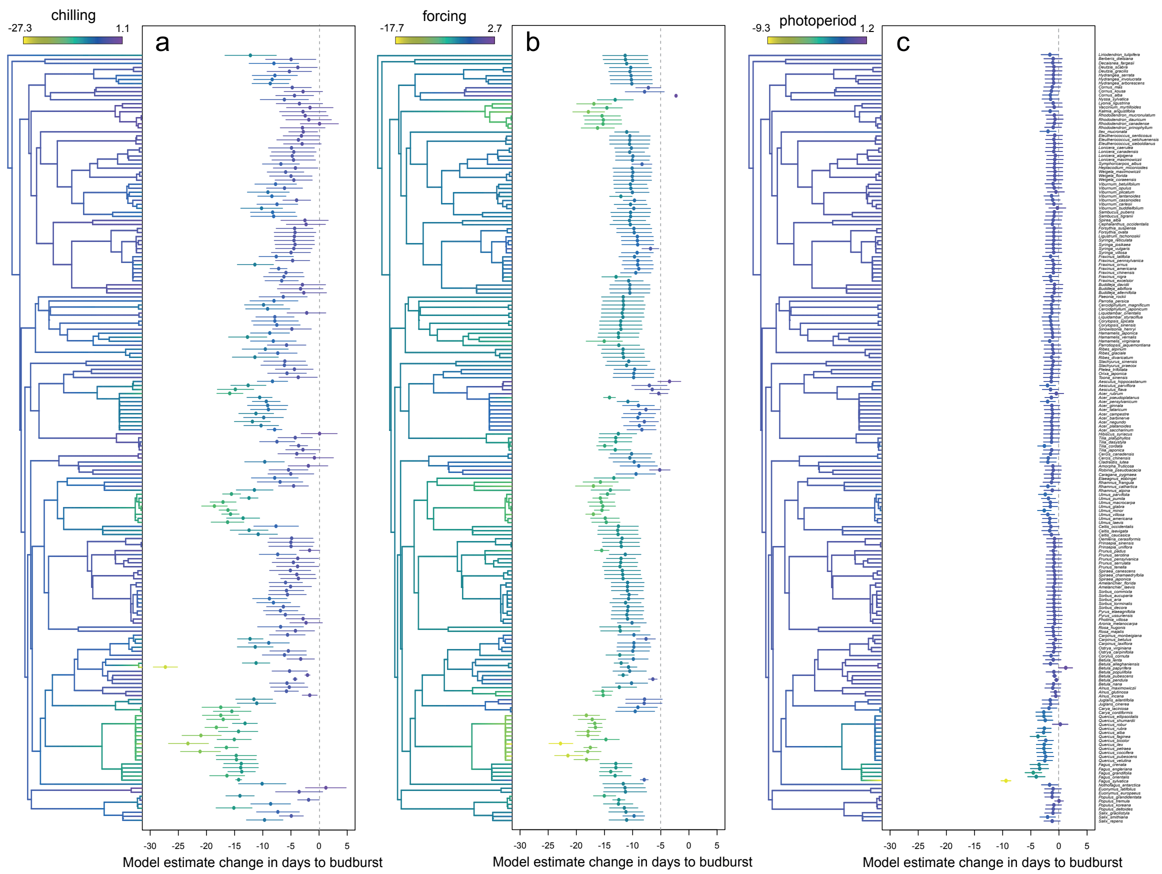

Figure 2: Estimated chilling, forcing and photoperiod responses (in units standardized by the mean and SD for each response variable) from 191 woody angiosperm species show most species shift earlier two days per standardized photoperiod unit, except for *Fagus* species with the well-studied *Fagus sylvatica* showing the strongest response. Figure reproduced from Morales-Castilla *et al.* (2024).